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CCDS 322 – Applied machine learning

Final project report

Project Title:

Smoker Classification Using Machine Learning Techniques on Insurance Data

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Introduction

Machine learning has become an important part of many industries and daily life applications. It is widely used in healthcare, finance, education, and business to help organizations analyze data and make better decisions. In the insurance industry, machine learning can help companies analyze customer information and predict insurance costs more accurately.

One of the common challenges in the insurance field is predicting medical insurance charges based on customer characteristics such as age, BMI, smoking habits, gender, and region. Since these factors affect insurance costs differently, prediction models can help companies better understand risks and improve pricing strategies.

In this project, machine learning techniques are applied to analyze an insurance dataset and predict medical insurance charges. Different machine learning algorithms are used and compared to identify the most suitable model for this task.

Project Goal

The main goal of this project is to compare different machine learning algorithms and identify the most suitable model to classify individuals as smokers or non-smokers based on their demographic and health data. By analyzing features such as age, BMI, and medical history, we aim to determine which variables have the strongest impact on smoker status and evaluate the classification performance of models like Logistic Regression, SVM, and Random Forest.

In this project, we used the Insurance Dataset from Kaggle because it contains several important features related to customer demographics and insurance expenses. By analyzing these factors, we aim to determine which variables have the strongest impact on medical insurance charges and evaluate the performance of different machine learning models.

Research Questions

This project aims to answer the following research questions:

- Can machine learning models accurately predict medical insurance charges?
- Which customer features have the greatest effect on insurance costs?
- Which machine learning algorithm performs best for predicting insurance charges?

Hypothesis

Our working theory is that a person's smoking status isn't random but is closely tied to specific health and demographic markers—most notably their medical expenses, BMI, and age. We expect to see that individuals with significantly higher medical charges are more likely to be classified as smokers.

We expect that machine learning models such as Random Forest and Lgistic Regression will achieve good prediction performance because they can identify relationships.

Tools and Libraries Used

The following tools and libraries were used in this project:

Python	Python was used as the main programming language for data analysis and machine learning model development.
Pandas	Used for loading the dataset, cleaning the data, handling missing values, and preprocessing.
NumPy	Used for numerical calculations and data manipulation.
Matplotlib	Used for visualizing data distributions and plotting important charts.
Scikit-learn	Used for train-test splitting, preprocessing, machine learning models, and evaluation metrics.
StandardScaler	Used for feature scaling to normalize the data and ensure all features have the same scale.
LabelEncoder	Used to transform categorical labels into numerical format for model processing.
Logistic Regression	Used as the classification algorithm to predict the target variable (Smoker).
Evaluation Metrics	Used to calculate Accuracy, Precision, and Recall to measure how well the model performs.
ListedColormap	Used to define the colors for visualizing the decision boundaries in the plots.

These tools were selected because they are reliable, efficient, and widely used in machine learning applications.

Dataset Description

In this section, we describe the dataset used in this project and explain why it is suitable for predicting medical insurance charges. The dataset contains customer information such as age, gender, BMI, smoking habits, region, and insurance charges. Initial exploration and preprocessing steps were performed before applying machine learning models.

Dataset Overview

The dataset used in this project is the Insurance Dataset obtained from Kaggle.

This dataset contains customer information related to medical insurance charges, including age, gender, BMI, number of children, smoking habits, and residential region. These features are important because they can affect insurance costs in different ways.

The dataset was selected because it provides real-world data and enough features to build machine learning models for predicting medical insurance charges.

The target variable selected for this project is (Smoker).

it showed two classes with very similar frequencies:

- Yes
- No

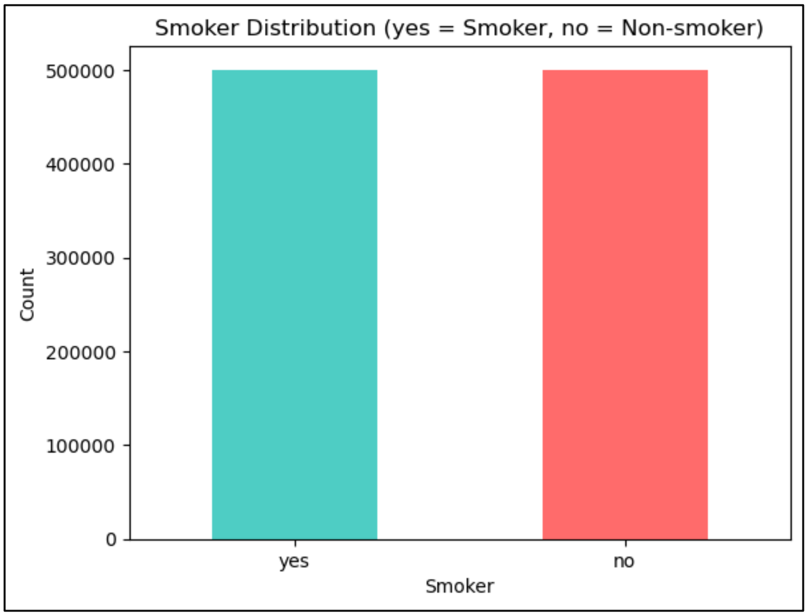

Figure 1, Distribution of Smoker Classes

This indicates that the dataset is balanced and suitable for classification tasks.

Important Features

All features in the dataset were used in this project because each variable may contribute to predicting the target variable (Smoker).

- Age
- Gender
- BMI
- Children
- Region
- Medical_History
- Family_Medical_History
- Exercise_Frequency
- Occupation
- Coverage_Level
- Charges
- Smoker (Target Variable)

No columns were removed during the feature selection process because all features were considered important and potentially useful for predicting the target variable (Smoker). Each variable provides different information related to customer health, lifestyle, and insurance details, which may help improve the performance of the machine learning models.

Data Exploration

Before we strated building machine learning models, we explored the dataset to better understand its structure and quality.

The following steps were performed during data exploration:

- Displaying the first few rows of the dataset
- Checking the dataset shape (number of rows and columns)
- Identifying column names and data types
- Checking for missing values
- Checking for duplicate rows
- Understanding the distribution of the target variable
- Visualizing Smoker distribution using a bar chart

The exploration showed that the dataset contains both numerical and categorical features, which require preprocessing before model training. It also showed that the dataset contains no duplicate values, while some missing values were found in only two columns before cleaning.

Data Preprocessing

In this section we will show the cleaning process of the data, and preprocessing steps that were applied to prepare the dataset for the machine learning models. These steps help improve the data quality, which increases the reliability and performance

Data Cleaning

After exploring the dataset, we found that the primary issue affecting data quality was a significant number of missing values, rather than outliers. Specifically, we identified **501,166 missing values** concentrated in the **medical_history** and **family_medical_history** columns.

To resolve this problem and prepare the data for our machine learning models, we handled the missing data based on its type: numerical columns were filled using the median value, while categorical columns were filled using the mode

```
============================================================
BEFORE CLEANING
============================================================

Dataset Shape: (1000000, 12)
Duplicate Rows: 0
Total Missing Values: 501166

Missing Values per Column:
  medical_history: 250,762
  family_medical_history: 250,404
```

Feature Encoding

To prepare the dataset for machine learning models, categorical features were transformed into numerical values using encoding techniques. This step was necessary because machine learning algorithms cannot process text-based categorical data directly.

- Target Variable Encoding: The target variable (Smoker) was encoded into binary values, where “yes” was represented as 1 and “no” was represented as 0.
- Categorical Features Encoding: LabelEncoder was applied to categorical features such as Gender, Region, Medical_History, Family_Medical_History, Occupation, and Coverage_Level to convert them into numerical values required for model training.
- Feature Preparation: After encoding, the dataset became fully numerical and ready for preprocessing, train-test splitting, and machine learning model implementation.

Train-Test Split

For the initial evaluation of the Logistic Regression model, we divided the dataset into two parts:

- **Training Set**: 80% of the data was used to train the model.
- **Testing Set**: 20% of the data was reserved for testing.
- **Stratification**: We applied the stratify=y parameter based on the binary target variable to ensure consistent class distribution and prevent bias in this specific model's results.

Models and Methodology

In this section, different machine learning models and techniques were applied to classify customers as smokers or non-smokers. The models were trained and evaluated using the prepared dataset after preprocessing, feature encoding, and train-test splitting. Model performance was evaluated using several metrics such as accuracy, precision, recall, and confusion matrix.

Logistic Regression

Logistic Regression was implemented as our baseline model for binary classification. The following steps and configurations were applied:

Feature Selection: The model was trained using BMI and Charges as the primary predictors to analyze their impact on smoking classification and visualize the decision boundary.

Model Configuration: We used the class_weight='balanced' parameter to automatically adjust weights according to class frequencies and improve classification performance between smokers and non-smokers.

Optimization: The model used random_state=0 and max_iter=1000 to ensure stable and reproducible results during training.

Feature Scaling: StandardScaler was applied to normalize the feature values before training the model.

Decision Boundary: The model established a linear decision boundary to separate smoker and non-smoker classes using BMI and Charges features.The figures show the Logistic Regression decision boundaries for both the training and testing datasets using BMI and Charges as the main features. The red and green regions represent the predicted classes generated by the model, while the points represent the actual data samples.

The model created a linear boundary that separates smokers and non-smokers based on the relationship between BMI and Charges. Most green points are concentrated in the upper region, while most red points appear in the lower region, showing that higher charges are more associated with smokers.

The training and testing plots show similar patterns, which indicates that the model generalized reasonably well and did not significantly overfit the training data.

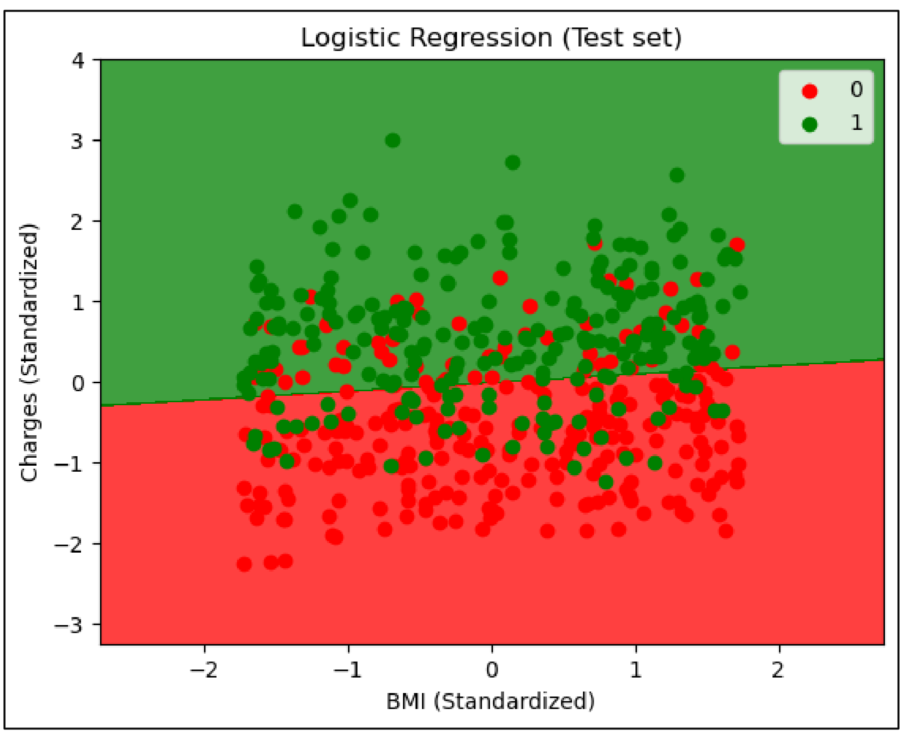

Figure 2, Logistic Regression (test set)

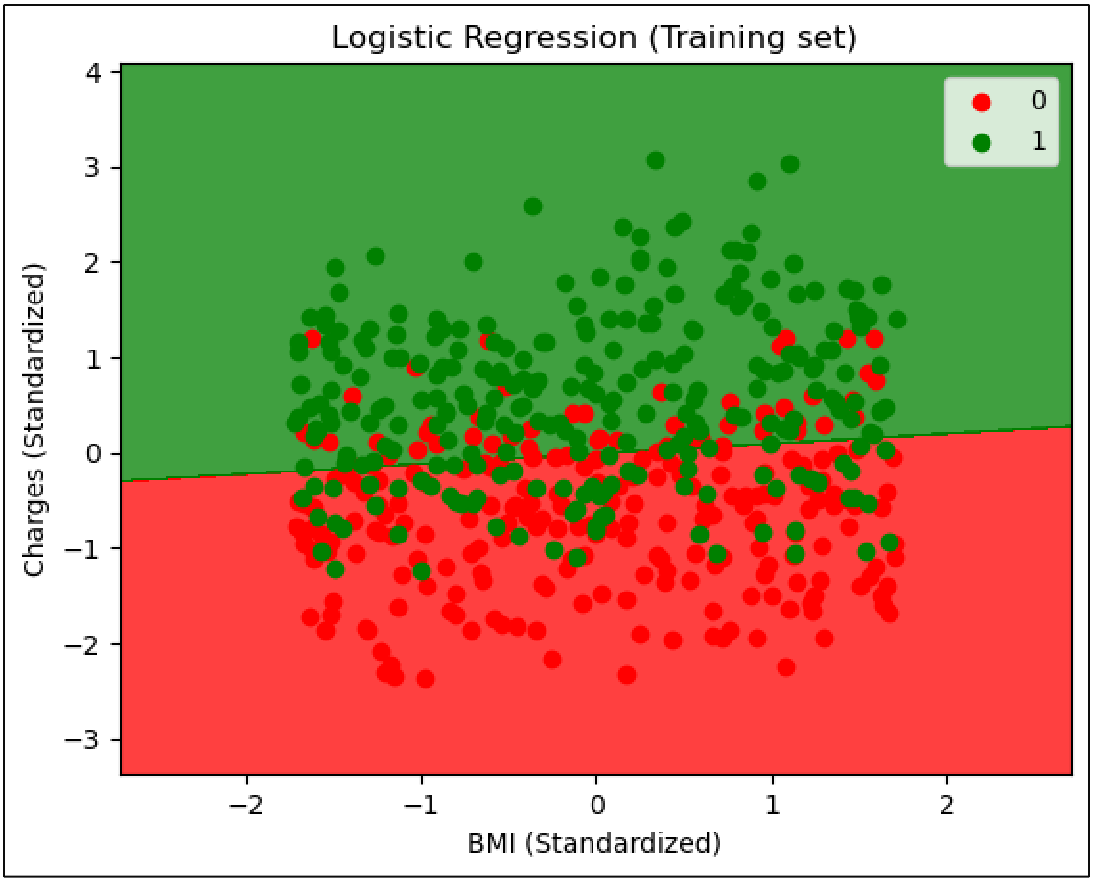

Figure 3, Logistic Regression (training set)

```
Confusion Matrix:
[[7750 2526]
 [2659 7637]]
```

Figure 4, Logistic Regression Matrix

Support Vector Machine (SVM)

Support Vector Machine (SVM) was implemented as another classification model to classify smokers and non-smokers based on the selected features. Different kernel functions were tested to compare their performance and identify the most suitable decision boundary for the dataset.

Linear

The Linear Kernel SVM model was applied to classify smokers and non-smokers using the selected features. The model created a linear decision boundary to separate the two classes based on the relationship between BMI and Charges. The model achieved an accuracy of 80.77%, indicating good classification performance. The visualization shows that most smoker samples were distributed in the upper region, while non-smoker samples appeared mostly in the lower region. However, some overlap between the two classes can still be observed near the decision boundary, which explains the presence of a few misclassified samples.

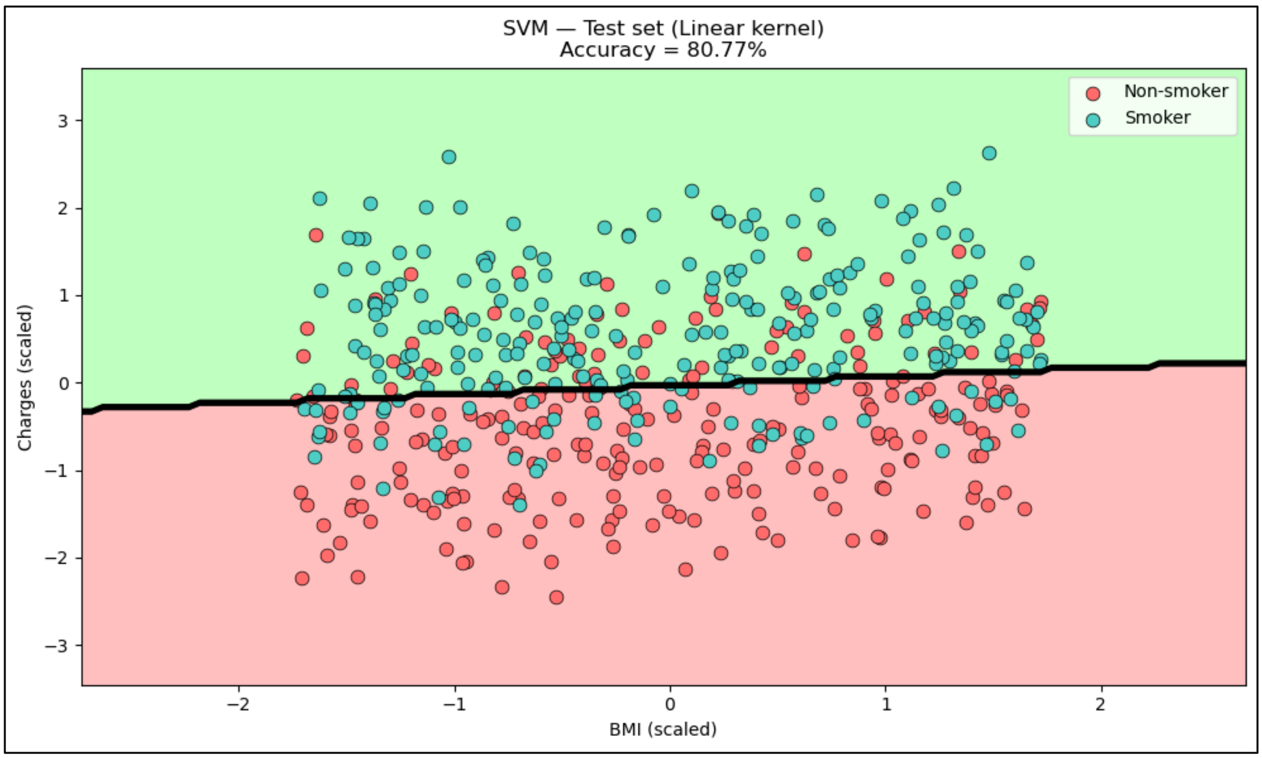

Figure 5, Linear Decision Boundary for Smoker Classification

```
Linear:
 [[1487  388]
 [ 333 1542]]
```

Figure 6, Linear Confusion Matrix

RBF

The RBF Kernel SVM model was applied to classify smokers and non-smokers using the selected features. Unlike the linear kernel, the RBF kernel creates a non-linear decision boundary, allowing the model to capture more complex relationships between BMI and Charges. The model achieved an accuracy of 80.24%, which is slightly lower than the Linear Kernel model. The confusion matrix showed that most samples were classified correctly, with only a limited number of misclassified cases. In the visualization, the curved boundary reflects the non-linear behavior of the RBF kernel. However, some overlap between smokers and non-smokers is still visible near the boundary area, which explains the remaining classification errors.

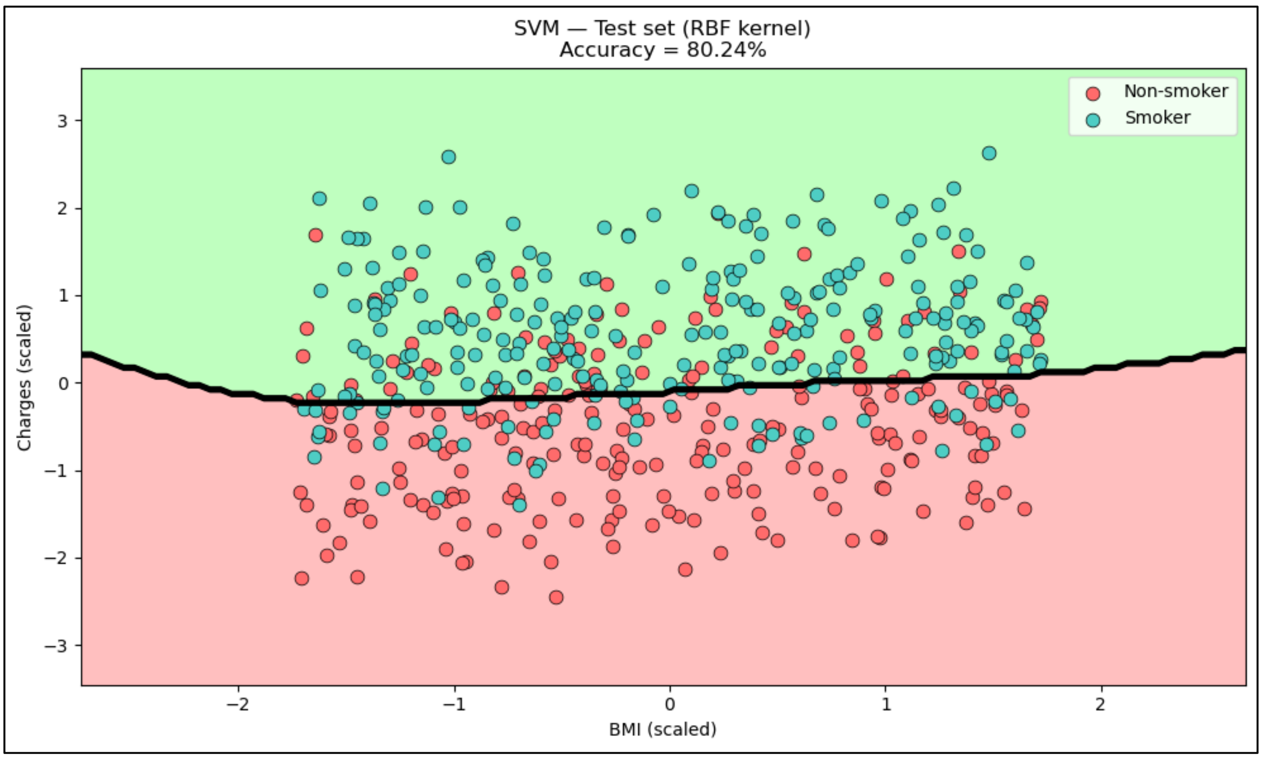

Figure 7, RBF Decision Boundary for Smoker Classification

```
RBF:
 [[1497  378]
 [ 363 1512]]
```

Figure 8, RBF confusion matrix

Polynomial

The Polynomial Kernel SVM model was applied to classify smokers and non-smokers using the selected features. This kernel creates a more flexible non-linear decision boundary compared to the Linear and RBF kernels, allowing the model to capture more complex patterns in the dataset. The model achieved the highest accuracy among all tested SVM kernels with an accuracy of 81.17%. The confusion matrix and classification report showed balanced performance between both classes, with precision, recall, and F1-scores remaining close across smokers and non-smokers. In the visualization, the polynomial decision boundary appears slightly curved, helping the model better separate overlapping samples. However, a small overlap between the two classes can still be observed near the boundary region, which resulted in a limited number of classification errors.

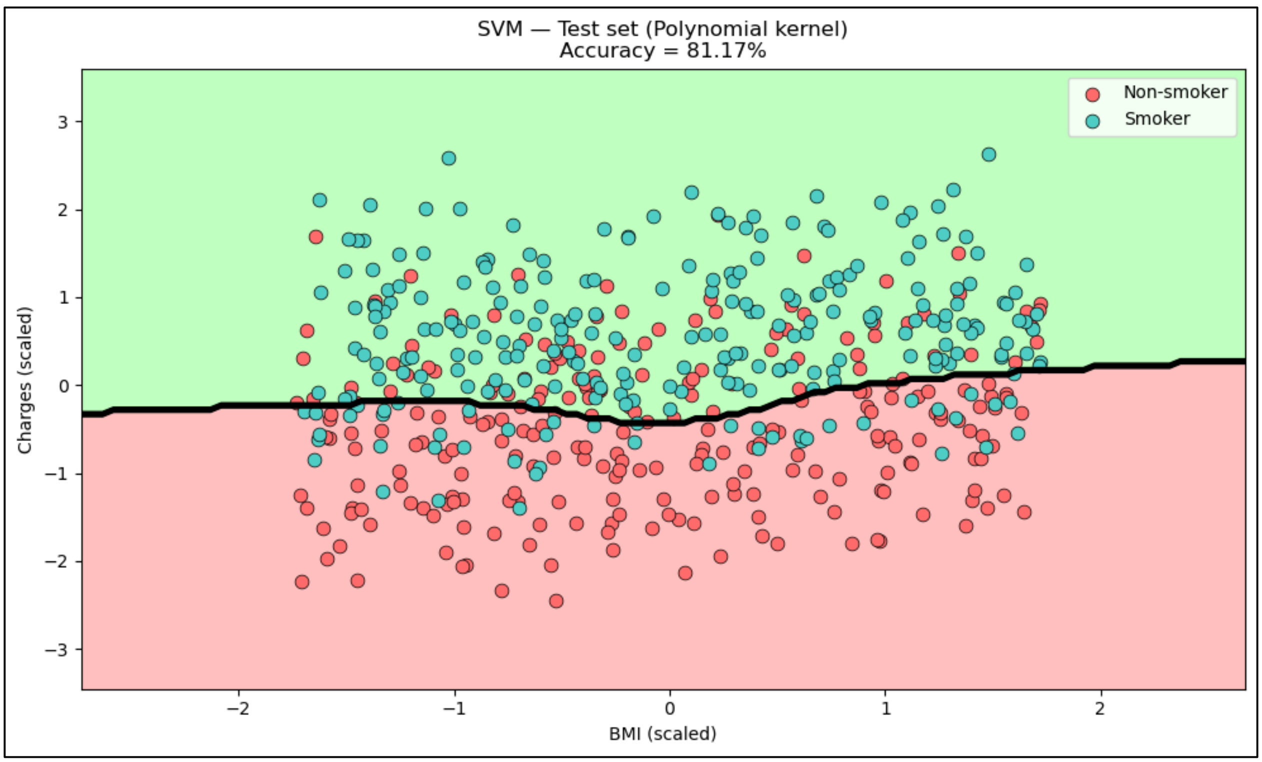

Figure 9, RBF Decision Boundary for Smoker Classification

```
Polynomial:
 [[1505  370]
 [ 336 1539]]
```

Figure 10, Polynomial Confusion matrix

Random Forest

Random Forest was implemented as the final classification model to predict whether a customer is a smoker or non-smoker. This model works by combining multiple decision trees to improve prediction performance and reduce overfitting. The dataset was split into 70% training and 30% testing, and all categorical features were converted using one-hot encoding before training.

The Random Forest model achieved an accuracy of 80.79%**,** which shows good classification performance. The classification report showed balanced results for both classes, with precision, recall, and F1-score around 0.81. The confusion matrix also showed that most samples were classified correctly, with some misclassified cases between smokers and non-smokers.

In addition, Random Forest provided feature importance results. The most important feature was charges, followed by BMI and age, indicating that these variables had the strongest influence on predicting smoking status. The decision boundary plot shows a more complex separation pattern compared to Logistic Regression and SVM, which is expected because Random Forest uses multiple decision trees.

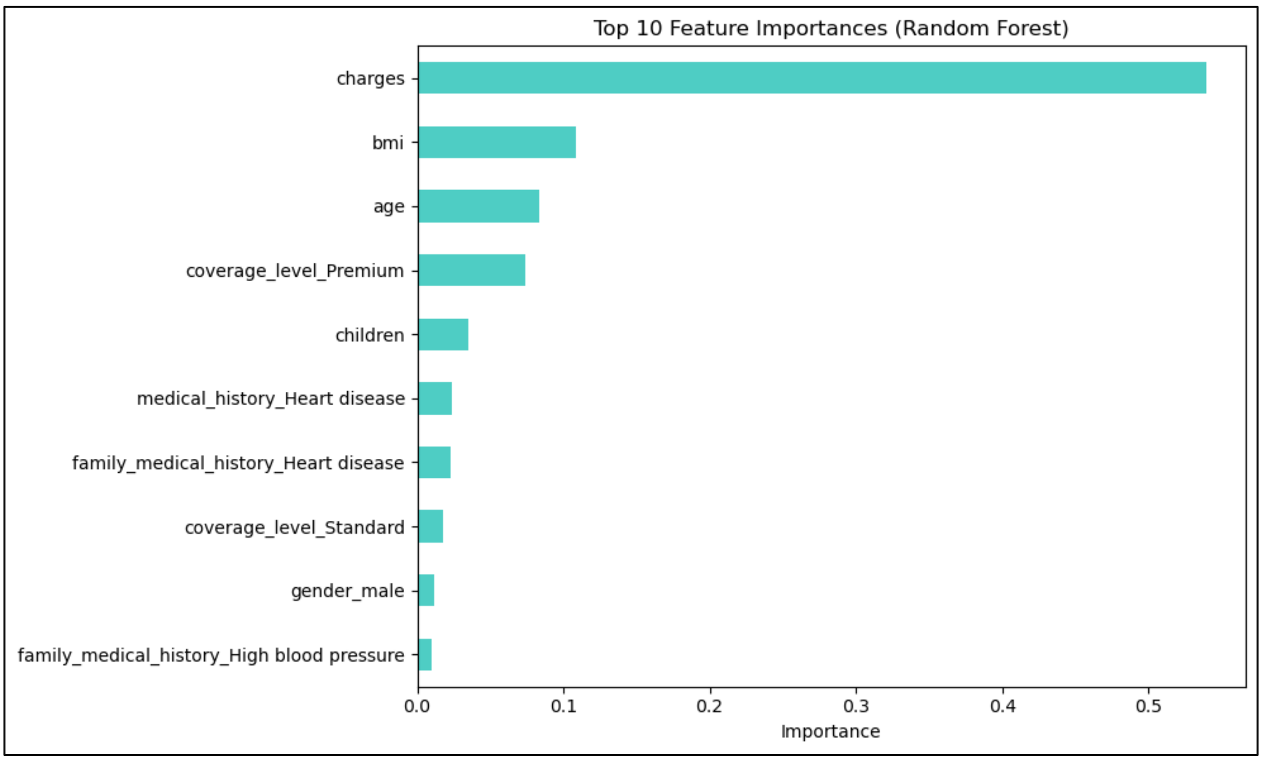

Figure 11, Top 10 Feature Importances in the Random Forest Mode

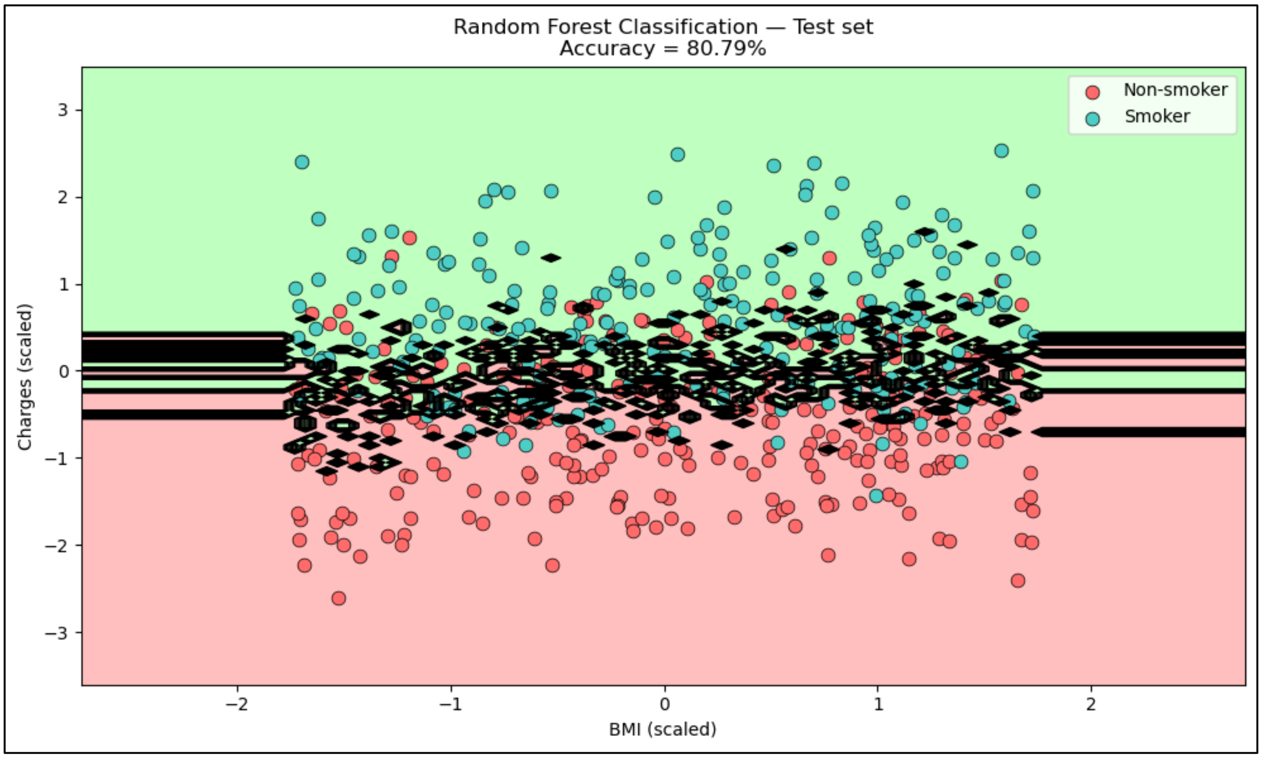

Figure 12, Random Forest Classification Decision Boundary on the Test Set

```
Confusion Matrix:
[[120416  29545]
 [ 28079 121960]]
```

Results and Evaluation

In this section, the performance of the implemented machine learning models was evaluated and compared using different evaluation metrics. The models were tested on unseen data to measure their ability to correctly classify smokers and non-smokers. Several metrics such as accuracy, precision, recall, F1-score, and confusion matrices were used to analyze the performance of each model and identify their strengths and limitations.

Evaluation Metrics

i. **Logistic Regression Results:**

- **Overall Accuracy:** The Logistic Regression model achieved an accuracy score of 74.77%, indicating that the model was able to correctly classify most smoker and non-smoker samples.
- **Precision and Recall:** The model achieved a precision score of 75.28 % and a recall score of 73.91 %, which shows balanced performance in identifying both classes with a moderate number of classification errors.

- **Confusion Matrix:** The confusion matrix showed 74,976 True Negatives and 73,986 True Positives, while there were 24,998 False Positives and 26,040 False Negatives, indicating some overlap between the two classes near the decision boundary.

ii. **Support Vector Machine (SVM)**

- **Overall Accuracy:** The SVM models achieved accuracy scores between 80.24% and 81.17%. The Polynomial kernel achieved the best performance with an accuracy of 81.17%.
- **Precision and Recall:** The models achieved balanced precision and recall scores around 0.80–0.82, showing stable classification performance.
- **Confusion Matrix:** Most samples were classified correctly, with fewer errors appearing near the decision boundary.
- **Decision Boundary Visualization:** The Polynomial kernel created a more flexible decision boundary compared to the Linear and RBF kernels, which slightly improved the model performance.

iii. **Random Forest**

- **Overall Accuracy:** The Random Forest model achieved an accuracy score of 80.79%, showing good classification performance.
- **Precision and Recall:** The model achieved 80.51% precision and recall 81.26%
- **Confusion Matrix:** The results showed a high number of correctly classified samples, with some classification errors still present.
- **Feature Importance:** The feature importance analysis showed that charges, BMI, and age were the most important features in predicting smokers.
- **Decision Boundary Visualization:** The visualization showed more complex and non-linear decision boundaries compared to Logistic Regression and SVM.

Model Comparison

The results showed that Random Forest and SVM outperformed Logistic Regression. While SVM achieved the highest raw accuracy at 81.17%, Random Forest delivered a strong and balanced performance with an accuracy of 80.79% and a better ability to handle complex, non-linear relationships within the data

Best Model Selection

Random Forest was selected as the best model for this project. Despite the marginal lead in raw accuracy for SVM, Random Forest is the superior practical choice due to its computational efficiency, interpretability, and robustness in handling large datasets and outliers

Ethical and Professional Responsibilities

When we are building these kinds of machine learning models that look at health habits and personal details, we have a big responsibility that goes way beyond just getting a high accuracy score or a good graph. Since these models can be used to make real decisions about people's money or their insurance coverage, we have to be really careful about how we handle the work as students and future professionals.

Data Privacy

The most important duty we have is protecting the privacy of the actual people behind the numbers in our dataset. Our file has a lot of personal info like medical history, age, and how much people pay for insurance. In a real professional job, we must make sure this data stays totally anonymous and is kept in a safe place. If this information leaked out because of a bad model or if we were careless with the data handling, it would be a huge breach of trust. We have an ethical job to make sure that someone's private health life doesn't become public information just because we were running an experiment.

Our Ethics

As data scientists, we also have to think a lot about being fair with our results. Machine learning models are only as good as the data we give them. If the data has some unfair patterns in it, the model will just "learn" those biases and repeat them. For example, if our model started penalizing a certain age group just because of a small trend it found, it could lead to those people paying higher insurance prices for no good reason at all. We have a professional responsibility to check our work and make sure the model isn't just following a stereotype.

Also, it is not ethical to just use a "black box" model and tell everyone that "the computer made the choice." If an insurance company is going to use a model like ours to decide if someone is a smoker or not, that person has a right to know exactly why that choice was made. We need to be able to explain how the model works and which things, like BMI or charges, are actually making the most impact on the result. At the end of the day, our work as developers should be to help solve problems in a way that is honest, transparent, and doesn't hurt anyone's basic rights.

Conclusion

In this project, several machine learning models were implemented and evaluated to classify smokers and non-smokers using the insurance dataset. The project included different stages such as data preprocessing, feature engineering, model training, visualization, and performance evaluation. Before training the models, the dataset was cleaned by handling missing values, removing duplicates, encoding categorical variables, and scaling numerical features to improve the overall quality of the data and ensure better model performance.

Three main machine learning approaches were tested in this project, including Logistic Regression, Support Vector Machine (SVM), and Random Forest. Different SVM kernels such as Linear, RBF, and Polynomial were also compared to analyze how different decision boundaries affect the classification performance. Several evaluation metrics including accuracy, precision, recall, F1-score, confusion matrices, and visualization plots were used to measure the effectiveness of each model and identify their strengths and weaknesses.

The results showed noticeable differences between the models. Logistic Regression achieved acceptable performance but struggled to correctly separate some smoker and non-smoker samples due to its linear decision boundary. On the other hand, SVM models achieved better and more stable classification results, especially when using the Polynomial kernel, which achieved the highest accuracy among all tested models. The Random Forest model also produced strong results and demonstrated the ability to capture more complex and non-linear patterns within the dataset. In addition, the feature importance analysis showed that variables such as charges, BMI, and age had the strongest influence on predicting smoker status.

The visualizations used throughout the project helped provide a clearer understanding of the data distribution and model behavior. The decision boundary plots showed how different algorithms classified the data and how some overlapping samples caused classification errors near the boundaries. These visualizations made it easier to compare the behavior of the models and interpret the obtained results.

Overall, this project demonstrated the importance of machine learning techniques in solving healthcare-related classification problems. It also highlighted the importance of preprocessing, feature selection, and choosing suitable algorithms to achieve better prediction performance. Through this project, practical experience was gained in implementing machine learning workflows, evaluating models, interpreting results, and visualizing data using Python and its related libraries.